(Please write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION
(B.Tech. IT/MAE/DMAM)
1st semester (January, 2023)
OFF LINE mode

Subject Code: BMA 110	Subject: Engineering Mechanics

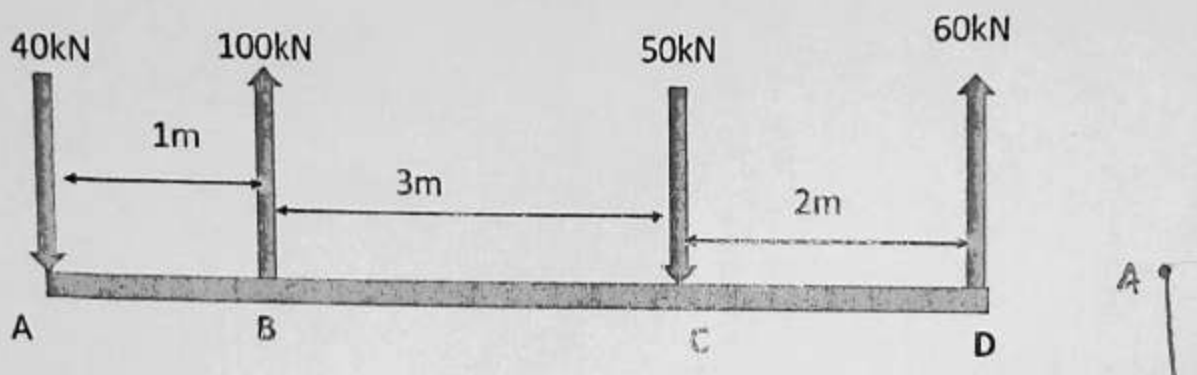
Time : 1 ½Hours Maximum Marks : 30

Q1 Attempt all Parts **(2.5*4)**

(a) what do you mean by non concurrent coplanar force system
(b) State Varignon's Theorem
c) State parallel Axis theorem and pappus theorem for area
(d) What is a truss. Write equations of equilibrium for a 2 dimensional system.

Q2 (Attempt any Two Parts) **(5,5)**

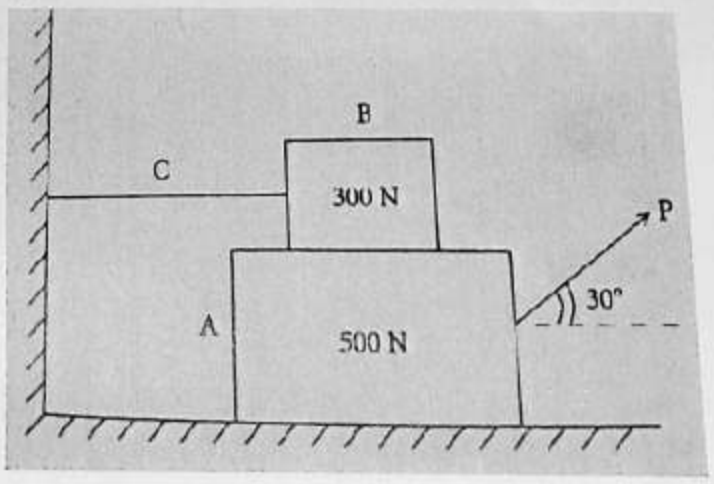
(a)A parallel coplanar force system is acting on a rigid bar as shown below





Reduce this force system into
a) a single force system and couple at B
b) A single force system

(b)For the system of blocks as shown in fig below find the force P required to move the lower block A and tension in the cable C . Take coefficient of friction for all contact surfaces to be 0.3

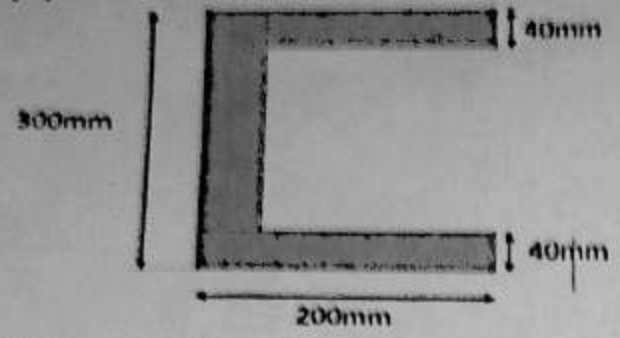


(C)For a ladder of length 4m, rest against a vertical wall making an angle of 45°, Determine the minimum horizontal force applied at A to prevent slipping if μ=0.2 between the wall and ladder and μ=0.3 for the floor and ladder . The ladder weight = 200N and a man weight 600N is at 3 m from point A (point A is on the floor and at the contact point of ladder and floor)

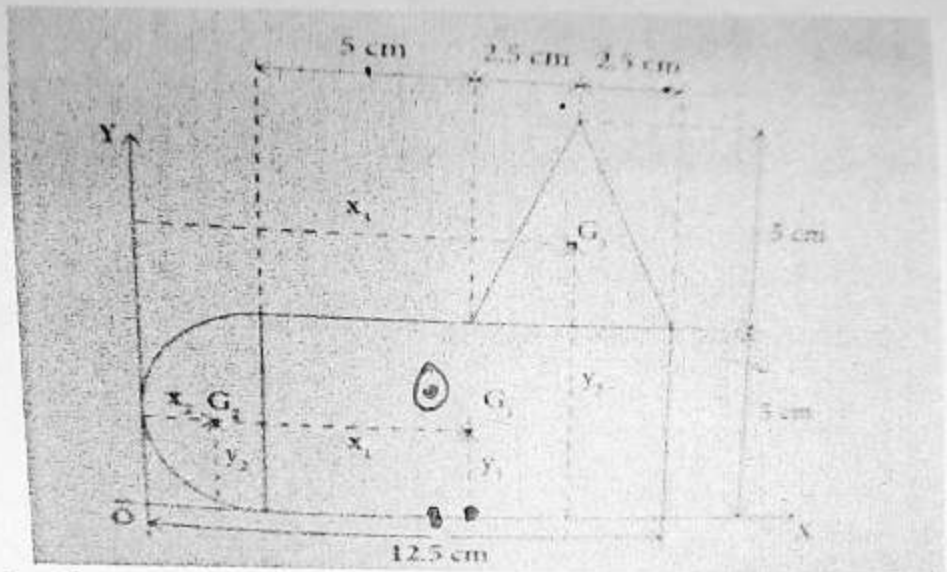
PTO

Q3 (Attempt any Two Parts **(5,5)**

(a)Find the MOI of C section about centroidal axis X-X



(b)Determine the centroid of the given figure



(C) Determine the forces in the truss members GE and CD of the given truss

